

RISHIKA AGARWAL IoT Assignment 2

Ques 1. I will choose ESP 32 for smart irrigation system because data is to be pushed to mobile app every 5 seconds. ESP has BLE / WiFi so it would be better for connecting to mobile to push data.

Ques 2. 500 sensors are across 2km:

LORA $\rightarrow$ 2-15km, low power,
WiFi $\rightarrow$ 50-100m, high power
BLE $\rightarrow$ 10-100m, low power
MQTT $\rightarrow$ app layer, depends.

$\hookleftarrow$ it is application layer messaging protocol.
We should use LoRa because it has range in 2km.

Ques 3. Problem with temperature sensor $\rightarrow$

- 1) Power Supply can be checked using multimeter.
- 2) Swap the sensor.
- 3) Check the conversion formula for temp.
- 4) Check for polarity and loose wiring.

Ques 4. To make ESP32 work as weather station on battery for 6 months we will choose Li-Po / LiFe battery first of all.

We sleep sensors to save power (when not needed).

Consider ESP-NOW which doesn't require WiFi or BT (reduces transceivers power).

Reduce CPU & speeds and work on an RT ~~time~~.

Ques 5 Automatic braking system in a vehicle requires ~~some~~ both almost zero delay. we would use CAN. we learned abt this in class.

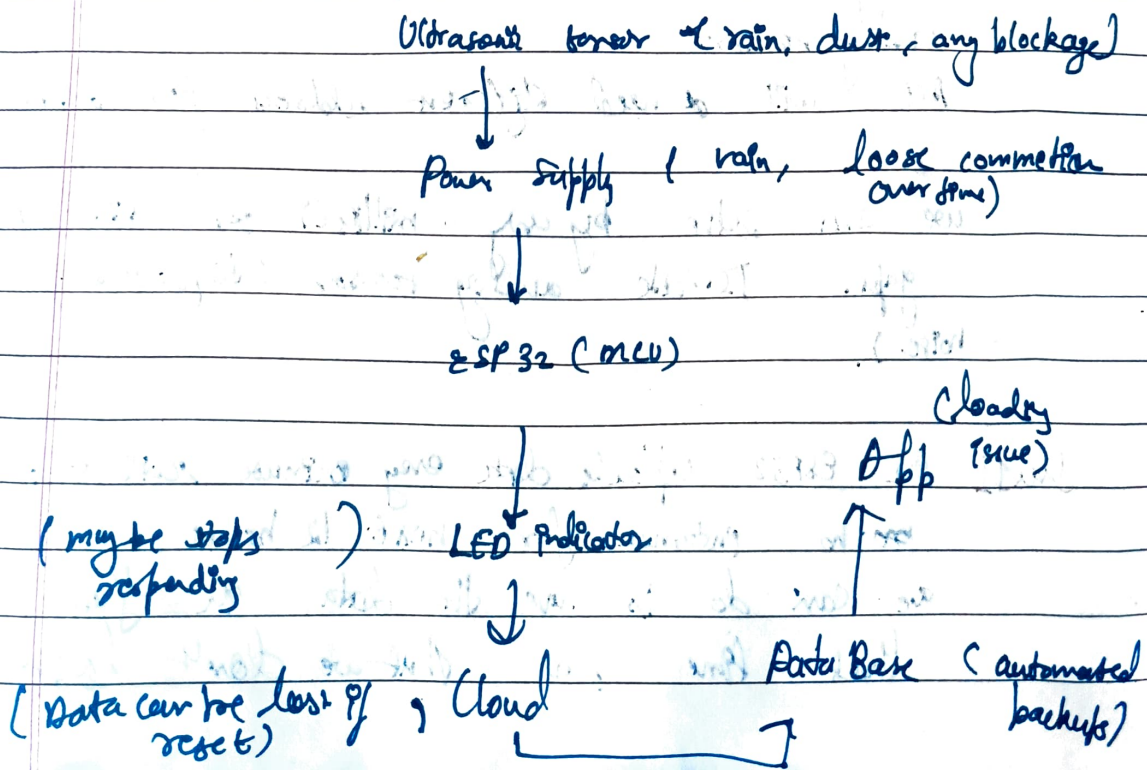
Allow multiple electronic devices to connect on a 2 wire bus.

Any node can ~~transmit~~ data ~~etc~~; dev can flexibly ~~transmit~~ data. good for automatic braking system.

Ques 6 Place the sensor in the middle of the room. make it away from windows, walls and doors. my equipment which detects ~~heat~~ etc.

Ques 7 Analog sensor can be better than digital sensors when we need ~~filteration~~ customisation when we need better resolution and ~~no~~ steps.

Ques 8 Parking Space Monitor:



Ques 1 Possible attacks on IoT device through wifi, case dropping, unauthorized command injection, jamming wpa compromised OTA channel.

5 Solutions:

- 1) TLS -- HTTPS and MQTT to use more.
- 2) Put IoT device on isolated network from personal devices.
- 3) WPA 2/WPA3 with unique phrase.
- 4) Unique passwords and usernames per device.
- 5) Device verifies a cryptographic signature before flashing. prevent malicious firmware injection.

Ques 10 we need to connect multiple sensors and actuators to the ESP32.

There will be conflict within GPIO pins. There will be conflict with time. Some of reading may be missed.

Delay can produce noise into I/O reading analog sensor.

All will need different address for communication.

we can solve by using millis() or use delay() gaps. Isolate analog sensor (to protect from noise).

Ques 11 An ESP32 uploads data every minute will receive no to internet for next 12 hours, so what we can do is save the data in flash for short time, so that we don't lose

Important readings

Ques 12 To make my IoT device work without human intervention we would like to avoid blockages (such as dust, moisture etc) on our IoT system. ~~Use~~ a powerfull energy saving battery.

For software part we can use sleep modes for battery saving purpose. Delays (for no possibility of system crash).

Ques 13 An IoT system can be smart without the use of Artificial Intelligence.

Smart means connected to a network. ~~not~~ but like a smart fan it connects to phone but isn't really any kind of AI.

Intelligent is something which learns and predicts output. which has a dataset to teach itself.

Smart is being able to connect to a network.

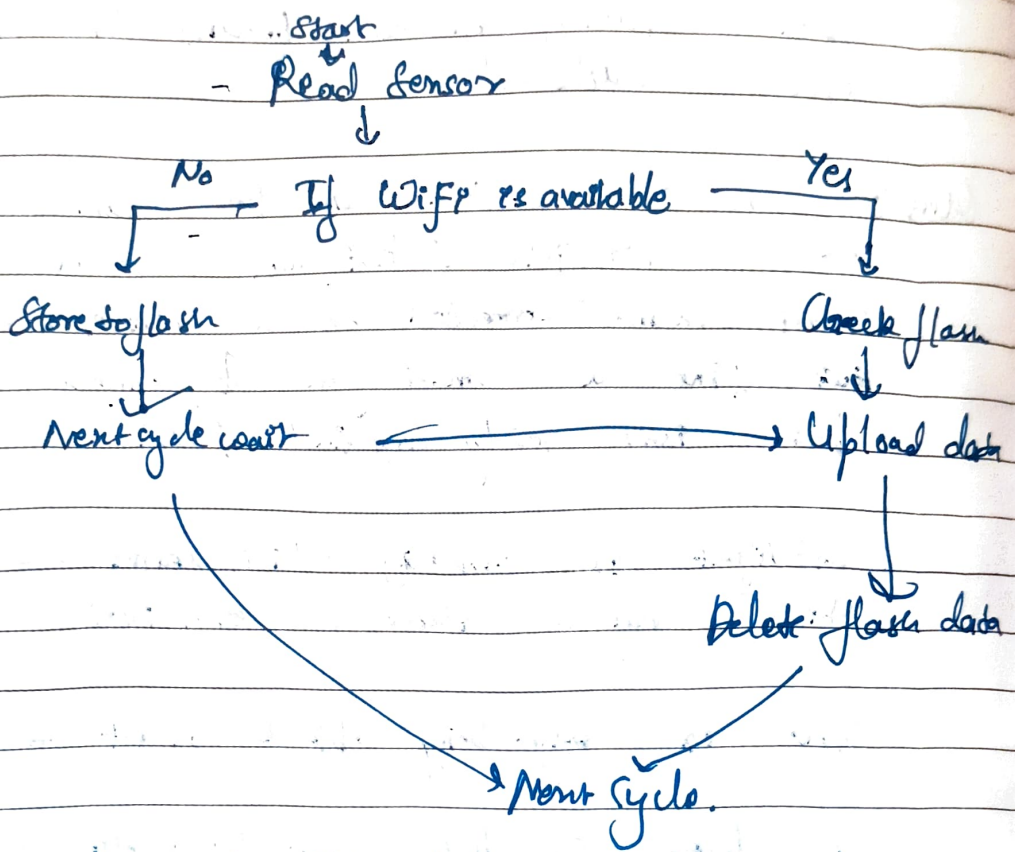
A device can be smart & intelligent.

Ques 14 Polling: The CPU repeatedly checks the ready at PIN in a loop. It helps in not missing or losing any data but also wastes CPU cycles.

Interrupts: The CPU is only reads when some event occurs. The CPU can sleep or do other work while no event takes place. Making the system optimized for battery saving.

Interrupts can be good for IoT systems when detection is to be done at certain occasions only and not continuously.

Ques 16 Flowchart for an IoT system that stores sensors readings locally whenever wifi is unavailable and automatically uploads data when connection is restored.



Ques 15 Trade offs in Gesture @ Combat game. →

- 1) Hard ware cost is high as 3 ESP32 are req'd which are communicating through ESP-NOW.
- 2) Responsiveness and delays of the gesture being accepted as a move in the game.

- 3) OLED only at centre, the 2 players may not be able to clearly view their moves and health bar.

Quest 7- ESP32 perform an OTA (over the Air) firmware update, including error handling if the update fails.

